An Additive Closure in the Chain of Prime Quadruplets*

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Abstract

The gap between consecutive prime quadruplets is a sum of at most eight distinct members of earlier quadruplets, with four small exceptions. Prime quadruplets are the constellations $(p, p+2, p+6, p+8)$ of four primes, the densest possible packing of four primes beyond $(5, 7, 11, 13)$. We order them into a chain $\mathcal{Q}_1 = (11, 13, 17, 19)$, $\mathcal{Q}_2 = (101, 103, 107, 109)$, $\mathcal{Q}_3 = (191, 193, 197, 199)$, ..., seeded by the *royal quad* $\mathcal{Q}_0 = (2, 3, 5, 7)$, and observe that the first member of every quadruplet can be written as the largest member of the previous quadruplet plus a sum of distinct members of earlier quadruplets, for example $2081 = 1879 + 199 + 3$ and $13001 = 9439 + 3461 + 101$; the other three members then follow without any search, as $p+2$, $p+(2\times 3)$ and $(p+6)+2$. The representation we record is fixed by one rule: after the base, always the largest earlier member that still leaves a solvable remainder, again and again, a royal value covering only what no member can, and a plain sum of royal members preferred over a product wherever any choice of members allows it. The precise form of the observation is that the gap D between a first member and its base is a sum of at most 8 distinct earlier members, with exactly four exceptional gaps, which we call the *royal gaps*, 22, 82, 172 and 382, that need a small expression in the royal members $2, 3, 5, 7$. Every other gap has a form with no royal value at all, which the utility and the app call its *no-royals* form. Members of earlier quadruplets are only ever added; $+$ and $\times$ act on nothing but the royal members, each used at most once and never more than two of them in a product. A congruence lemma modulo 210 shows that these four are the only exceptions in the entire admissible range below 10^9, not merely the ones we met. Allowing one such expression in every gap brings the bound down to six terms with no exception at all. Under a uniform form of the Hardy–Littlewood conjecture for eight linear forms we prove that every sufficiently large integer in the five residue classes modulo 210 that a gap can occupy is a sum of two first members, so that two terms eventually suffice. That hypothesis, and with it the asymptotic content of the conjecture, is an instance of the generalised Hardy–Littlewood conjecture of Green and Tao for systems of linear forms in primes; the system contains the pattern $(0, 2, 6, 8)$, so it has infinite complexity in their sense and lies outside their theorem, as does every statement about a fixed prime constellation. We therefore do not present the asymptotic statement as new. What is new is the explicit finite structure, the four royal gaps, the bound of eight, the quadruplet from which a Bertrand-type doubling holds, and a numerical test of the prediction itself: for every admissible integer below 10^7 we count the representations as a sum of two first members and compare the count with the singular-series prediction. Decade by decade the ratio of observed to predicted is 0.82, 0.79, 0.88 and 1.12; the count of quadruplets itself stands at 0.86, 0.89, 0.96 and 1.06 of its asymptotic over the same range, and after dividing out the square of that deviation the ratio is 1.11, 0.99, 0.96 and 1.01. The prediction is thus confirmed, at the first scale at

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which it is of order one, for a system that no theorem reaches. We also prove that the doubling, if it holds, forces the largest-first rule to take at most one member from each quadruplet beyond the third and at most $\log_2 D + 4$ members in all. We verify the closure conjecture for all 28,387 prime quadruplets below 10^9 with an independent checker, and the royal gaps and the doubling for all 180,528 below 10^{10}. Two further representations of each gap, one that prefers multiplication and one that prefers powers of chain members, are also computed and released. We test the objection that the pattern is a consequence of density rather than of primality by running the same solver on control chains of identical spacing whose members are not prime: the bound of eight and the parity of the term counts are reproduced by a control that preserves the residue class, and are therefore structural, but no control reproduces the small closed set of exceptions. We also prove, from a two-line covering lemma and computed base cases, that the chain is an additive basis for all integers: every integer up to the next quadruplet (indeed up to a bound that grows like the sum of all earlier members) is a sum of distinct chain members plus one expression in $2, 3, 5, 7$ in which, unlike in the conjecture itself, $-$ is allowed alongside $+$ and $\times$; this is unconditional below 10^9 and rests beyond that only on a mild growth condition. We explain carefully what the conjecture does and does not say about locating the next quadruplet, give a heuristic for why such representations are plentiful, and state open questions. We also make explicit the decimal consequences of the first-member congruence: every first member ends in the digit 1, and both the member itself and its digit sum have the form $2 + 3n$ for nonnegative integers n, properties verified for all 28,387 first members below 10^9. The utility that builds the chain and the full dataset are available at https://github.com/SPAlgorithm/LE.

1 Introduction

A *prime quadruplet* is a quadruple of primes of the form

$$(p,\ p+2,\ p+6,\ p+8),$$

the tightest admissible pattern of four primes. Apart from $(5, 7, 11, 13)$, every prime quadruplet has $p \equiv 11 \pmod{30}$, so its members end in the digits $1, 3, 7, 9$ in that order. The sequence of first members $11, 101, 191, 821, 1481, 1871, 2081, \ldots$ is OEIS A007530 [3]. Whether there are infinitely many prime quadruplets is open; it is the case $k = 4$ of the Hardy–Littlewood prime k-tuple conjecture [1], and the Bateman–Horn heuristic [2] predicts that the number of quadruplets up to x is asymptotically $C_4\, x/(\log x)^4$ with $C_4 \approx 4.15$. The strongest unconditional results in this direction are the bounded-gap theorems of Zhang [4] and Maynard [5], which give infinitely many pairs (and m-tuples) of primes within a bounded distance but not the specific pattern $(0, 2, 6, 8)$. Large quadruplets and related clusters have been searched for computationally for decades [6].

This note records an additive structure that runs through the whole sequence of prime quadruplets, at least as far as we have looked. Order the quadruplets by size and call the k-th one $\mathcal{Q}_k$; put the four primes $2, 3, 5, 7$ in front as $\mathcal{Q}_0$, the *royal quad*. Then the first member of $\mathcal{Q}_k$ is obtained from the largest member of $\mathcal{Q}_{k-1}$ by adding a few distinct members of earlier quadruplets, and the other three members of $\mathcal{Q}_k$ are that first member plus 2, plus 2×3, and plus $2 \times 3 + 2$. In one line:

The gap between consecutive prime quadruplets is a sum of at most eight distinct members of earlier quadruplets, with four small exceptions.

Distinctness carries the content: with repetition allowed the sentence would be empty, since every integer is a sum of copies of 2 and 3. Writing D for the gap between the first member of

$\mathcal{Q}_k$ and the largest member of $\mathcal{Q}_{k-1}$, a few cases read

$$\begin{aligned}
\mathcal{Q}_4 &: \quad 821 - 199 = 622 = 197 + 193 + 191 + 17 + 13 + 11,\\
\mathcal{Q}_5 &: \quad 1481 - 829 = 652 = 197 + 191 + 103 + 101 + 19 + 17 + 13 + 11,\\
\mathcal{Q}_{10} &: \quad 5651 - 3469 = 2182 = 2081 + 101,\\
\mathcal{Q}_{22} &: \quad 31721 - 25309 = 6412 = 3257 + 1483 + 1481 + 191,\\
\mathcal{Q}_{173} &: \quad 1022501 - 1022389 = 112 = 101 + 11,
\end{aligned}$$

each member used at most once and none of $2, 3, 5, 7$ among them. Exactly four gaps below 10^9 have no such sum, the *royal gaps* 22, 82, 172 and 382; they are small gaps with too few members below them (for $\mathcal{Q}_2$, $D = 101 - 19 = 82$, and the only earlier members $11, 13, 17, 19$ sum to 60) and are closed with a small expression in $2, 3, 5, 7$ (Section 3, Appendix B). Some representation by a plain sum of earlier members, with at most one added royal value, exists for every first member below 10^9 except 11 and the 30 first members whose gap is 82, which need one product of royal members, as in $101 = 19 + 17 + 13 + 11 + (5 \times 7) + (2 \times 3)$ (Section 6). The representation the utility stores, chosen by the rule of Definition 5, is a literal sum, of members alone or with one added royal value, for all but 30 of the 28,386 gaps, the occurrences of the gap 82, which close with one royal product.

Seen from additive number theory, the set of first members is an explicit and natural thin set of primes, of counting function about $C_4\, x/(\log x)^4$, and the conjecture says that it is an additive basis of bounded order for the residue classes it can occupy, with the basis property witnessed by an explicit certificate for every element. Wirsing [7] showed that the primes contain thin sub-bases of every order $h \geq 3$, but by an existence argument; we know of no explicit natural set of comparable sparsity for which a basis property of order two has been tested numerically at this scale, although the property itself is, as we explain next, an instance of a general conjecture. Section 8 proves it under a uniform form of the Hardy–Littlewood conjecture (Theorem 1) and tests the prediction against the data.

Relation to the general conjectures. The asymptotic content of the conjecture is not new, and we want to be exact about where it comes from. The circle method of Hardy and Littlewood predicts the number of solutions of any system of linear equations in primes; Green and Tao [8] state the prediction in full generality as their Conjecture 1.2, the *generalised Hardy–Littlewood conjecture*, for any finite system of affine-linear forms, and prove it for systems of *finite complexity*, those in which no two forms have proportional linear parts, the proof being completed by the inverse theorem of Green, Tao and Ziegler [9]. Writing an integer n as a sum of m first members of quadruplets is the system of the $4m$ forms p_i, $p_i + 2$, $p_i + 6$, $p_i + 8$ $(1 \leq i < m)$ and $n - \sum_{i<m} p_i + \{0, 2, 6, 8\}$ in $m - 1$ variables. It contains pairs of forms with the same linear part, p_1 and $p_1 + 2$ for instance, and so has infinite complexity for every m, exactly as the twin prime conjecture has. Conjecture 1.2 nevertheless predicts, for every fixed $m \geq 2$ and every n in the admissible residue classes, a number of representations that tends to infinity with n, and therefore predicts that all but finitely many gaps are sums of two first members; Theorem 1 is this case $m = 2$ written out with its singular series. The same conjecture, for the single system $(p, p + 2, p + 6, p + 8)$, predicts a quadruplet between t and $2t$ for all large t. The reader should therefore regard Conjectures 1 to 3 and the doubling of Remark 3 as instances of a conjecture already in the literature, and far out of reach of proof: no single pattern $(0, 2k)$ is known to contain infinitely many prime pairs, despite [4, 5], and the nearest theorem to our statement is that of Matomäki and Shao [10], that every large odd integer is a sum of three primes each of which has another prime at a bounded distance; for a single prime, $m = 1$ in the notation above, the statement is Vinogradov's three primes theorem [11], completed by Helfgott [12]. We are grateful to a number theorist who pointed this out to us.

What the general conjecture does not give is the finite part: which small gaps fail, how many terms suffice in practice, and from which quadruplet on the doubling holds. Those are what this

note computes. They are statements of the same kind as Dubner's conjecture that every even number above 4208 is a sum of two primes each having a twin [13], or Richert's theorem that every integer above 6 is a sum of distinct primes [14] and Brown's criterion for such completeness [15]: an explicit exceptional set, found by computation, in front of an asymptotic statement that is known or expected. We also test the asymptotic prediction itself, which the general conjecture invites and which, to our knowledge, has not been done for this system: Section 8 compares the predicted and the observed number of two-term representations for every admissible integer below 10^7 and for every gap below 10^{10}.

Section 3 states this precisely in three forms of increasing strength, Section 4 proves a companion statement, that with subtraction allowed among the royal members every integer below the next quadruplet is such a sum, Section 5 walks through examples that can be checked by hand, Section 6 reports the computation up to 10^9, and Section 8 discusses, with some care, what the statement means for finding the next quadruplet and proves, under the Hardy–Littlewood conjecture, that two terms eventually suffice.

The observation grew out of the authors' earlier work on additive decompositions of primes, *Ladhe's Conjecture* [16], which concerns single primes written as sums of three primes and underlies a hash-based signature scheme [17]. The present note is about prime *quadruplets* and is self-contained.

Proposition (Last digit and digit sum). *Let $(p, p+2, p+6, p+8)$ be a prime quadruplet with $p \geq 11$, and let $s_{10}(p)$ denote the sum of the decimal digits of p. Then*

(i) $p \equiv 1 \pmod{10}$*: the first member ends in the digit* 1*, and the four members end in* $1, 3, 7, 9$*;*

(ii) $p \equiv 2 \pmod{3}$*, that is,* $p = 2 + 3m$ *for some integer* $m \geq 3$*;*

(iii) $s_{10}(p) \equiv 2 \pmod{3}$*, that is,* $s_{10}(p) = 2 + 3n$ *for some* $n \in \mathbb{Z}_{\geq 0}$*.*

Proof. The first-member congruence $p \equiv 11 \pmod{30}$ gives $p \equiv 1 \pmod{10}$ and $p \equiv 2 \pmod{3}$, which are (i) and (ii); the other members $p+2$, $p+6$, $p+8$ are then $3, 7, 9 \pmod{10}$. Write $p = \sum_{j=0}^{r} a_j 10^j$, where a_j are its decimal digits. Since $10 \equiv 1 \pmod{3}$, we have $s_{10}(p) = \sum_{j=0}^{r} a_j \equiv p \equiv 2 \pmod{3}$. The digit sum is positive, so it is $2 + 3n$ with $n \geq 0$; this is (iii). □

For example, $101 = 2 + 3 \cdot 33$ with $s_{10}(101) = 2$, $191 = 2 + 3 \cdot 63$ with $s_{10}(191) = 11$, and $821 = 2 + 3 \cdot 273$ with $s_{10}(821) = 11$. This is an unconditional consequence of the existing congruence, valid at any size, rather than a further conjecture limited to the computed range. It concerns the first member of each $\mathcal{Q}_k$, $k \geq 1$; the royal seed and $(5, 7, 11, 13)$ are excluded. The condition is necessary but not sufficient: 131 ends in 1 and has digit sum 5, but $133 = 7 \times 19$ is composite, so 131 does not begin a prime quadruplet.

2 Definitions and rules

Definition 1 (Quads)**.** $\mathcal{Q}_0 = (2, 3, 5, 7)$ *is the* royal quad*. For $k \geq 1$, $\mathcal{Q}_k$ is the k-th prime quadruplet $(p, p+2, p+6, p+8)$ with $p \geq 11$, in increasing order of p. We write its members as $q_k^{(1)} = p$, $q_k^{(3)} = p+2$, $q_k^{(7)} = p+6$, $q_k^{(9)} = p+8$, the superscript being the last decimal digit. The quadruplet $(5, 7, 11, 13)$ is not part of the chain, because* 5 *and* 7 *already belong to $\mathcal{Q}_0$, and $\mathcal{Q}_0$ itself is not of the form $(p, p+2, p+6, p+8)$.*

Definition 2 (Base, gap, pool)**.** *For $k \geq 2$ the* base *of $\mathcal{Q}_k$ is $m_k = q_{k-1}^{(9)}$, the largest member of the previous quadruplet. For a member t of $\mathcal{Q}_k$ the* gap *is $D = t - m_k > 0$. The* pool *available to $\mathcal{Q}_k$ is $\mathcal{P}_k = \mathcal{Q}_1 \cup \cdots \cup \mathcal{Q}_{k-1}$, the set of all $4(k-1)$ primes of the earlier quadruplets.*

Definition 3 (Royal expressions)**.** *A* royal expression *is an arithmetic expression built from the members* $2, 3, 5, 7$ *of* $\mathcal{Q}_0$*, each used at most once, with the operations* $+$ *and* $\times$ *only, where a product may multiply only* two *members:* $(5 \times 7) + (2 \times 3)$ *is a royal expression,* $2 \times 5 \times 7$ *and* $5 \times (7 + (2 \times 3))$ *are not. Let* $\mathcal{R}$ *be the set of its values;* $\mathcal{R}$ *has* 30 *elements between* 2 *and* 41*, and every one of them is* flat *(no product contains a sum). One subset matters below:* $\mathcal{R}_+$*, the* 11 *values reachable by addition alone,* $\{2, 3, 5, 7, 8, 9, 10, 12, 14, 15, 17\}$.

Definition 4 (Additive representation)**.** *An* additive representation *of a member* t *of* $\mathcal{Q}_k$ *(*$k \geq 2$*) is an identity*

$$t \;=\; m_k \;+\; s_1 + s_2 + \cdots + s_j \;+\; r, \qquad s_1 > s_2 > \cdots > s_j, \;\; s_i \in \mathcal{P}_k, \;\; r \in \mathcal{R} \cup \{0\},$$

with all s_i *distinct and none equal to* m_k*. Members of earlier quadruplets are thus only ever added; multiplication occurs only inside the royal expression* r*, and only between two of* $2, 3, 5, 7$*. Its* length *is* j *if* $r = 0$ *and* $j + 1$ *otherwise. Several* s_i *may come from the same quadruplet. (The stricter reading "at most one member per quadruplet" already fails at* $\mathcal{Q}_2$*:* $101 - 19 = 82$*, and no single member of* $\mathcal{Q}_1$ *leaves a remainder in* $\mathcal{R}$*.) A member* t *of* $\mathcal{Q}_1$ *is written as a royal expression alone, e.g.* $11 = (2 \times 3) + 5$.

When all four members of one quadruplet occur among the summands we abbreviate them by the largest member followed by two dashes: `19--` *stands for* $19 + 17 + 13 + 11$*,* `199--` *for* $199 + 197 + 193 + 191$*, and* `7--` *for the whole royal quad* $7 + 5 + 3 + 2$*. This is a display convention only: the four members remain four distinct terms, the length of the representation is unchanged, and the dataset stores every term. The utility prints this form by default and spells every term out under* `--full`*.*

Only the first member p *of* $\mathcal{Q}_k$ *is given such a representation by search. Its three companions are written*

$$p + 2 \;=\; p + 2, \qquad p + 6 \;=\; p + (2 \times 3), \qquad p + 8 \;=\; (p + 6) + 2,$$

which is all the quadruplet pattern requires of them; they are neither searched for nor listed. From here on "the gap" of $\mathcal{Q}_k$ *means the gap* $D = p_k - m_k$ *of its first member.*

Definition 5 (Stored representation)**.** *Among the additive representations of a first member the utility stores the one given by the following rule. Start with* $r = D$*. While* $r > 0$ *and some unused member of* $\mathcal{P}_k$ *does not exceed* r*, take the largest such member and continue with the remainder; when no unused member fits, the remainder must be* 0 *or an* allowed *royal value, which closes the equation; when it is neither, or a choice leads to a remainder that cannot be finished, the next smaller member is tried instead (the search backs up). The allowed closings are widened in three passes, and the first pass that succeeds gives the stored form:*

(i) 0 *or a plain sum of royal members (*$\mathcal{R}_+$*:* 3*,* $5 + 3$*,* $7 + 5 + 2$*, . . .);*

(ii) additionally the values with one product of two royal members ($(3 \times 5) + 7$*,* $(5 \times 7) + 3$*, . . .);*

(iii) additionally the two values that need two products.

Below 10^9 *pass (i) succeeds for every gap but* 82*, whose two equations (*101 *and the shared one) come from passes (iii) and (ii). The terms* $s_1 > s_2 > \ldots$ *thus come out largest first, a royal value covers only what no member can, and a product is written only where every choice of members would need one. This is exactly the closing order of the utility's* `-d` *switch (Section 4), so the stored equation and the derivation of the same number agree whenever the latter closes with a royal sum. For the shared equation of a gap the base is among the candidates (any quadruplet reusing the gap has its base above it); a first member whose base lies below its own gap, which happens for* 101 *and* 821 *only, keeps an equation of its own with the base excluded. The bounds of Section 3 concern the* shortest *representations, which a separate search computes (Section 6); the stored form is usually longer.*

The following congruences are elementary but shape everything below.

Lemma 1. *Let $\mathcal{Q}_k = (p, p+2, p+6, p+8)$ be a prime quadruplet with $p \geq 11$ and write $M = p+4$, so that its members are $M-4$, $M-2$, $M+2$, $M+4$. Then*

(i) $M \equiv 15 \pmod{30}$*, equivalently* $p \equiv 11 \pmod{30}$*;*

(ii) $M \not\equiv \pm 2, \pm 4 \pmod 7$*, hence* $M \equiv 15, 105, 195 \pmod{210}$ *and*

$$p \equiv 11,\ 101,\ 191 \pmod{210},$$

equivalently $p \equiv 2,\ 3,\ 4 \pmod 7$*;*

(iii) consecutive first members differ by $h \equiv 0,\ 30,\ 90,\ 120,\ 180 \pmod{210}$*;*

(iv) the gaps of the four members of $\mathcal{Q}_k$ satisfy

$$D \equiv 22,\ 24,\ 28,\ 0 \pmod{30} \quad \textit{for } t = q_k^{(1)}, q_k^{(3)}, q_k^{(7)}, q_k^{(9)} \textit{ respectively,}$$

and, more precisely, D lies in one of the twenty classes

$$D \equiv 0,\ 22,\ 24,\ 28,\ 30,\ 82,\ 84,\ 88,\ 90,\ 112,\ 114,\ 118,\ 120,\ 172,\ 174,\ 178,\ 180,\ 202,\ 204,\ 208 \pmod{210}.$$

Since only the first member is searched for, the gaps that actually occur are those of $t = q_k^{(1)}$: $D \equiv 22 \pmod{30}$, in one of the five classes $22, 82, 112, 172, 202 \pmod{210}$, the five classes 22, 82, 112, 172, 202 being those of a first member;

(v) every gap is even while every pool member is odd, so a representation with $r = 0$ has an even number of terms.

Proof. All four members are primes larger than 7, hence coprime to $2, 3, 5, 7$. Modulo 3 they are $M+2$, $M+1$, $M+2$, $M+1$, so $M \equiv 0$; modulo 5 they are $M+1$, $M+3$, $M+2$, $M+4$, so $M \equiv 0$; and M is odd. This is (i). Modulo 7 none of $M \pm 2$, $M \pm 4$ may vanish, so $M \equiv 0, \pm 1$, and combining with (i) leaves the three classes of (ii). Part (iii) is the set of differences of $\{11, 101, 191\}$ modulo 210. For (iv), writing $\delta \in \{0, 2, 6, 8\}$ for the offset of t inside $\mathcal{Q}_k$,

$$D = (p_k + \delta) - (p_{k-1} + 8) \equiv h + \delta - 8 \pmod{210},$$

and the four values of δ give the four classes modulo 30, the five values of h from (iii) the twenty classes modulo 210. Finally D is a difference of two odd numbers, hence even, and a sum of j odd numbers is congruent to j modulo 2, which is (v). □

Pool members are $\equiv 11, 13, 17, 19 \pmod{30}$, hence a two-term representation of the gap of a first member ($D \equiv 22$) must add two first members ($11 + 11 \equiv 22$).

Three ways. The utility described in Section 6 computes, for every member, one representation of each of three kinds:

A *additive*, as in the definition above, chosen so that addition is exhausted before any royal multiplication is used (fewer multiplications first, then fewer terms, then larger members first);

M *multiplicative*: any two distinct members of $\mathcal{P}_k \cup \mathcal{Q}_0$ may be multiplied, and multiplication is preferred over addition (the largest unused member is multiplied by the largest member that still fits, then the next, plain addition only when nothing can be multiplied);

E *exponential*: in addition, a member may be raised to a member power, powers being preferred over products over plain addition. Because the order in which powers are tried changes the outcome, two variants are kept: E1 takes the largest base first, E2 the largest power value first;

M1 *without the base*: the M search applied to the member itself instead of to its gap. No base term is required, the pool is every member of $\mathcal{P}_k \cup \mathcal{Q}_0$ below the target, and the largest products carry the number, e.g. $854921 = (427249 \cdot 2) + (109 \cdot 3) + (17 \cdot 5) + 11$;

E3 *powers first*: the largest power b^e below the member, with b and e distinct members, then again the largest power below what is left as long as it covers at least half of it, followed by the remainder in the additive way (members below the target added, largest first, closed by a flat royal expression), e.g. $854921 = 829^2 + 11^5 + 17^3 + 1489 + 107 + 101 + 19$ and $19497221 = 11^7 + 2^{13} + 829 + 827 + 199 + 3$. If no choice of members closes the remainder, the chain of powers is shortened by one, and finally the next largest first power is tried.

In A, M, E1 and E2 every member of $\mathcal{P}_k \cup \mathcal{Q}_0$ is used at most once and the base m_k is always a plain term; M1 and E3 also use every member at most once, the base and exponent of the E3 power included, but have no base term.

3 The conjecture

We state the observation in three forms. The first is the one the utility works with; the second and third are the ones we consider interesting, because they bound the number of terms.

Conjecture 1 (existence form)**.** *The first member p_k of every prime quadruplet $\mathcal{Q}_k$ with $k \geq 2$ has an additive representation over the chain: $p_k = m_k + \sum_i s_i + r$ with distinct $s_i \in \mathcal{P}_k$ and $r \in \mathcal{R} \cup \{0\}$; the other three members then follow as $p_k + 2$, $p_k + (2 \times 3)$ and $(p_k + 6) + 2$.*

Conjecture 2 (bounded pure form)**.** *There is an absolute constant K such that, apart from finitely many gaps, every gap $D = p_k - m_k$ ($k \geq 2$) is a sum of at most K distinct members of $\mathcal{P}_k$. The data support $K = 8$, with exactly four exceptional gaps, which we call the* royal gaps*:*

$$D \in \{22,\ 82,\ 172,\ 382\},$$

each of which needs a royal value. No new *exceptional gap occurs beyond $\mathcal{Q}_{435}$ (first member* 3,512,231*) up to 10^9, and the list is in fact complete in a stronger sense than observation: by Lemma 1 every gap lies in one of five classes modulo* 210*, and among all* 9,523 *even numbers in those classes up to* 400,000*—a range that covers every gap below 10^9, the largest being* 395,512*—exactly these four are not sums of at most eight distinct chain members below themselves. Four further small values,* 52*,* 262*,* 472 *and* 772*, are equally unreachable but can never occur as gaps, their classes being excluded by the lemma. The last occurrence of an exceptional gap below 10^9 is at $\mathcal{Q}_{28063}$ (first member* 985,124,051*). In all,* 98 *of the* 28,386 *gaps are exceptional. A separate run of the utility to 10^{10} (*180,528 *quadruplets,* 180,527 *gaps, the largest* 846,052*) finds the same four royal gaps,* 348 *occurrences in all, and no fifth.*

No-royals forms. A sum of pool members alone, with no royal value, is what the utility (`lc.py`, switch `--no-royals`) and the companion app call the *no-royals* form of a gap. Conjecture 2 says that every gap except 22, 82, 172 and 382 has one of at most eight terms, even though the stored equation, which closes largest first with a royal value whenever one fits, usually shows a royal value. For $\mathcal{Q}_{30}$ the stored equation is $72221 = 69499+2089+199+197+193+19+17+5+3$ and the shortest no-royals form is $72221 = 69499 + 1877 + 821 + 13 + 11$; for $\mathcal{Q}_{15}$, $16061 = 15739 + 199 + 109 + 11 + 3$ becomes $16061 = 15739 + 197 + 101 + 13 + 11$. The shortest such form, with the larger members first among forms of the same length, is listed under the stored equation of every first member in Appendix A; `none` marks the four exceptional gaps. Below 10^9, 23,749 of the 28,386 stored first-member equations close with a royal value although a no-royals form exists, 4,539 are already pure, and the remaining 98 are the occurrences of the four exceptional gaps.

Conjecture 3 (bounded form with one flat royal value)**.** *Every gap $D = p_k - m_k$ ($k \geq 2$) is a sum of at most five distinct members of $\mathcal{P}_k$ plus at most one element of $\mathcal{R}$; that is, every first member has an additive representation of length at most six.*

Remark 1 (Why the existence form alone is weak)**.** *Once a few quadruplets exist, the pool is large and the sums of its subsets cover every even number in range many times over, so Conjecture 1 is very likely to hold for reasons that have little to do with quadruplets; it would also be vacuous if only finitely many quadruplets existed. What is not automatic is that a* fixed *number of terms suffices while the gaps grow: the average gap between consecutive quadruplets near x grows like $(\log x)^4/C_4$, about* 35,000 *at* 10^9*, and the pool members that fit under a gap are few (*92 *below the average gap and* 336 *below the largest gap,* 395,520*, in our data). Conjectures 2 and 3 would be falsified by a single gap that is not a short sum; the natural falsification test is a very large gap between consecutive quadruplets.*

Remark 2 (The royal gaps)**.** *All four exceptions are small and appear early. The gap* 82 *belongs to $\mathcal{Q}_2$ (and reappears whenever two consecutive quadruplets are* 90 *apart, as at $\mathcal{Q}_3$, $\mathcal{Q}_{14}$, $\mathcal{Q}_{437}$, ...); at that point the pool is $\mathcal{Q}_1$ alone, whose subset sums cannot reach* 82*. The gaps* 22*,* 172 *and* 382 *are coverage holes of $\mathcal{Q}_1 \cup \mathcal{Q}_2$: for instance $382 = 1871 - 1489$ is reached by no subset of the members below it; its stored form $199+109+19+17+13+11+7+5+2$ closes with the royal sum* 14*. With the* 11 *royal addition values allowed, only* 82 *remains exceptional; with all of $\mathcal{R}$ allowed there is no exception. In the light of Lemma 1 the four are exactly the coverage holes that survive the congruence filter: of the eight values below* 400,000 *that no sum of at most eight distinct members reaches, four lie in classes no gap can occupy. Within the admissible classes the pressure relaxes quickly: below* 400,000 *six terms suffice for every value above* 1552*, and four for every value above* 53,362*. Appendix B lists the four gaps with their forced representations.*

4 Coverage: every integer below the next quadruplet

The chain also serves as an additive basis for *all* integers, not only for the quadruplets, once subtraction, and products of any shape, are allowed inside the royal part. The two settings should not be confused: the conjecture writes each quadruplet with $+$ and $\times$ only, at most two royal members to a product; the derivation of an arbitrary integer below also uses $-$, and its royal part may be nested. This section proves the second statement from two computed base cases.

Definition 6 (Signed royal values)**.** *$\mathcal{R}^{\pm}$ is the set of non-negative values of expressions built from the members $2, 3, 5, 7$ of $\mathcal{Q}_0$, each used at most once, with $+$, $-$ and $\times$ placed anywhere; unlike $\mathcal{R}$, there is no restriction to two-factor products. It has* 83 *elements between* 0 *and* 210*, contains every integer from* 0 *to* 38*, and its first gaps are $39, 43, 53, 59, 61, 62$. Examples: $0 = 5 - (3 + 2)$, $1 = 3 - 2$, $4 = 7 - 3$, $33 = (5 \times 7) - 2$, $67 = (2 \times 5 \times 7) - 3$. The utility reaches for $\mathcal{R}$ first and for the signed values only when no choice of primes allows a closing in $\mathcal{R}$ ($928 = 823 + 103 + 2$ rather than $829 + 19 + 13 + (2 \times 5 \times 7) - 3$); the values with three-factor or nested products are the last resort of all. Below* 300,000 *that resort is taken for exactly one integer: $99 = 19 + 13 + 67$ has no derivation with two-factor products, since every remainder $99 - s$ over subsets s of $\{11, 13, 17, 19\}$ misses the two-factor values. A nested product (a sum inside a product, such as $5 \times ((2 \times 7) - 3)$) is never used.*

Definition 7 (Derivation)**.** *A* derivation *of an integer $n \geq 0$ over $\mathcal{Q}_0, \ldots, \mathcal{Q}_k$ is an identity $n = s_1 + \cdots + s_j + r$ with distinct $s_i \in \mathcal{Q}_1 \cup \cdots \cup \mathcal{Q}_k$ and $r \in \mathcal{R}^{\pm}$ ($j = 0$ or $r = 0$ allowed). Quadruplet members are only added; subtraction and multiplication occur only among royal members. Write C_k for the set of integers that have a derivation over $\mathcal{Q}_0, \ldots, \mathcal{Q}_k$ and T_k for the largest T with $[0, T] \subseteq C_k$.*

By direct computation,

$$T_0 = 38, \qquad T_1 = 156, \qquad T_2 = 576, \qquad T_3 = 1356, \qquad T_4 = 4656,$$

so with the pool $2, 3, 5, 7, 11, 13, 17, 19$ every integer from 0 to 156 has a derivation, in particular every integer up to 101; for instance $16 = 11{+}5$, $53 = 19{+}11{+}(7{\times}5){-}2$ and $55 = 19{+}11{+}(7{\times}5)$.

Lemma 2. *Let A be a set of non-negative integers containing every integer in $[0, T]$, and let $0 < m \le T + 1$. Then $A \cup (A + m)$ contains every integer in $[0, T + m]$.*

Proof. An integer $x \in [0, T]$ lies in A. An integer $x \in [T + 1, T + m]$ satisfies $0 \le x - m \le T$, so $x - m \in A$ and $x \in A + m$. □

Proposition 1. *Put $S_3 = T_3 = 1356$ and $S_k = S_{k-1} + 4p_k + 16$ for $k \ge 4$. If $p_j \le S_{j-1} + 1$ for every j with $4 \le j \le k + 1$, then $T_k \ge S_k \ge p_{k+1}$: every integer n with $0 \le n \le p_{k+1}$ has a derivation over $\mathcal{Q}_0, \ldots, \mathcal{Q}_k$. For $k \le 3$ the same conclusion holds by the computed values ($T_0 \ge p_1 = 11$, $T_1 \ge p_2 = 101$, $T_2 \ge p_3 = 191$, $T_3 \ge p_4 = 821$).*

Proof. Induction on $k \ge 3$ with the claim $T_k \ge S_k$; the case $k = 3$ is the computation. Assume $T_{k-1} \ge S_{k-1}$ and add the members p_k, $p_k + 2$, $p_k + 6$, $p_k + 8$ of $\mathcal{Q}_k$ one at a time, each with coefficient 0 or 1. The first satisfies $p_k \le S_{k-1} + 1 \le T_{k-1} + 1$ by hypothesis, so Lemma 2 raises the coverage to $T_{k-1} + p_k$; each following member is at most $p_k + 8$, far below the coverage reached so far, so the lemma applies three more times and $T_k \ge T_{k-1} + 4p_k + 16 \ge S_k$. Finally $p_{k+1} \le S_k + 1$ and p_{k+1} is odd while S_k is even, hence $p_{k+1} \le S_k$. □

The hypothesis is mild: it fails only if a quadruplet's first member exceeds $S_{j-1} + 1$, which grows like four times the sum of all earlier first members. In the chain up to 10^9 it holds at every $j \le 28{,}387$, and it is tightest at $j = 4$ ($p_4 = 821$ against $S_3 + 1 = 1357$). The one-step condition $p_j \le 5p_{j-1} + 16$ is sufficient and also holds throughout (the largest ratio p_j/p_{j-1} for $j \ge 3$ is 4.30, at $j = 4$). So Proposition 1 is unconditional below 10^9; an unconditional proof for all k would follow from any Bertrand-type bound for prime quadruplets, which is open, although far more than what is needed here.

Remark 3 (A Bertrand-type doubling, and whole quadruplets in a representation)**.** *Let $t_k = p_k + 8$ be the top member of $\mathcal{Q}_k$. Bertrand's postulate puts a prime between n and $2n$; the data show the same for quadruplets once the chain is under way (Table 1). For every $k \ge 4$ the whole of $\mathcal{Q}_{k+1}$ lies between the top member of $\mathcal{Q}_k$ and its double,*

$$t_k < p_{k+1} < t_{k+1} < 2t_k \qquad (k \ge 4),$$

and the same holds at $k = 2$ ($199 < 218$), while the royal quad has only its first member below the double ($11 < 14 < 19$). The only failures in the whole chain are $k = 1$ ($p_2 = 101 > 38$) and $k = 3$ ($p_4 = 821 > 398$). For $k \ge 4$ the ratio t_{k+1}/t_k never exceeds 1.796 (at $k = 4$, 1489/829). This was checked for all 28,386 consecutive pairs below 10^9 and, in a separate run of the utility, for all 180,527 pairs below 10^{10}.

The rule of Section 2 takes the largest unused member not above the remainder, so it takes t_k only when the remainder lies below the smallest unused member above t_k; while $\mathcal{Q}_{k+1}$ is untouched that member is p_{k+1}, and a second member of $\mathcal{Q}_k$ can then follow only if $p_{k+1} \ge 2t_k - 2$. The doubling statement therefore confines whole quadruplets to the start of the chain, and the data agree: below 10^{10} no representation uses two members of the same quadruplet beyond $\mathcal{Q}_3$, $\mathcal{Q}_2$ contributes at most three of its four members, and a quadruplet folds whole, in the notation `19--` of Section 2, only for $\mathcal{Q}_0$, $\mathcal{Q}_1$ and $\mathcal{Q}_3$. Below 10^9, `7--` occurs in 789 representations, `19--` in 3,161, `199--` in 644, and `19--` together with `7--` in a further 51 (below 10^{10}: 6,074, 19,005, 3,994 and 502); the counts fold the base with its quadruplet, as the display does. Whether $p_{k+1} < 2t_k$ holds for every $k \ge 4$ is a Bertrand-type question for quadruplets, open like the growth condition above, although under the Hardy–Littlewood heuristic the ratio tends to 1.

primes: n to $2n$			quadruplets: top member t_k to $2t_k$				
n	$2n$	primes between		t_k	$2t_k$	how many	first members between
2	4	3	$\mathcal{Q}_0$ (royal)	7	14	1	11
3	6	5	$\mathcal{Q}_1$	19	38	**0**	none
5	10	7	$\mathcal{Q}_2$	109	218	1	191
7	14	11, 13	$\mathcal{Q}_3$	199	398	**0**	none
11	22	13, 17, 19	$\mathcal{Q}_4$	829	1,658	1	1,481
13	26	17, 19, 23	$\mathcal{Q}_5$	1,489	2,978	2	1,871, 2,081
			$\mathcal{Q}_6$	1,879	3,758	3	2,081, 3,251, 3,461
			$\mathcal{Q}_9$	3,469	6,938	1	5,651
			$\mathcal{Q}_{10}$	5,659	11,318	1	9,431
			$\mathcal{Q}_{20}$	22,279	44,558	4	25,301, 31,721, ...
			$\mathcal{Q}_{100}$	477,019	954,038	59	490,571, 495,611, ...
			$\mathcal{Q}_{1000}$	11,740,039	23,480,078	640	11,765,051, ...
			$\mathcal{Q}_{10000}$	265,227,499	530,454,998	7,080	265,381,211, ...
			$\mathcal{Q}_{100000}$	4,825,866,979	9,651,733,958	75,469	4,825,896,611, ...

Table 1: Bertrand's postulate for primes, and its quadruplet analogue read off the chain: the quadruplets whose first member lies strictly between the top member $t_k = p_k + 8$ of $\mathcal{Q}_k$ and its double. The analogue fails at $\mathcal{Q}_1$ and $\mathcal{Q}_3$ and nowhere else below 10^{10}: all 102,946 intervals with $k \geq 4$ and $2t_k \leq 10^{10}$ contain a quadruplet, exactly one only at $k = 4, 9, 10$, exactly two at $k = 5, 7, 8$, and at least ten from $k = 24$ on. Beyond 10^{10} the question is open.

The second paragraph of Remark 3 can be made exact. It is the one place in this note where the doubling statement implies something about the representations, and the implication is unconditional once the doubling is assumed.

Proposition 2 (one member per quadruplet, under doubling)**.** *Assume $p_{k+1} < 2t_k$ for every $k \geq 4$. Start from any integer $D \geq 1$ and run the largest-first process over the members of $\mathcal{Q}_1, \mathcal{Q}_2, \ldots$: while some unused member is at most the remainder, subtract the largest such member. Then*

(i) no two members of the same quadruplet $\mathcal{Q}_j$ with $j \geq 4$ are subtracted;

(ii) if the member subtracted belongs to $\mathcal{Q}_j$ with $j \geq 4$ and the remainder before it is r, the remainder after it is below $r/2 + 8$;

(iii) at most $\log_2\big((D + 16)/837\big) + 1$ members of $\mathcal{Q}_4, \mathcal{Q}_5, \ldots$ are subtracted, hence fewer than $\log_2 D + 4$ members in all when $D \geq 821$.

In particular no quadruplet beyond $\mathcal{Q}_3$ ever appears whole, and the process spends at most one term on each quadruplet beyond $\mathcal{Q}_3$.

Proof. Write $m_1 > m_2 > \cdots$ for the members subtracted, in order, and $r_0 = D$, $r_i = r_{i-1} - m_i$ for the remainders; the m_i decrease because m_{i+1} is an unused member not above $r_i \leq r_{i-1}$ and m_i was the largest such member. The remainders decrease as well. We first show:

$(*)$ *If $m_i \in \mathcal{Q}_j$ with $j \geq 4$ and $r_{i-1} \geq p_{j+1}$, then all four members of $\mathcal{Q}_{j+1}$ are among $m_1, \ldots, m_{i-1}$.*

Indeed $p_{j+1} > t_j \geq m_i$ and $p_{j+1} \leq r_{i-1}$, and m_i is the largest unused member not above r_{i-1}; so p_{j+1} is used, $p_{j+1} = m_h$ with $h < i$. Then $r_h \geq r_{i-1} \geq p_{j+1}$, so the remainder before m_h was $r_{h-1} = r_h + p_{j+1} \geq 2p_{j+1} > p_{j+1} + 8$. The members $p_{j+1} + 2$, $p_{j+1} + 6$, $p_{j+1} + 8$ lie in $(m_h, r_{h-1}]$, and m_h was the largest unused member not above r_{h-1}; so they were used before m_h. This proves $(*)$.

(i) Suppose two members of some $\mathcal{Q}_j$, $j \geq 4$, are subtracted, and let m_i be the first of them. The second, m_g with $g > i$, satisfies $p_j \leq m_g \leq r_{g-1} \leq r_i$, so $r_{i-1} = r_i + m_i \geq p_j + m_i \geq 2p_j$.

The hypothesis gives $p_{j+1} < 2t_j = 2p_j + 16$, and $p_{j+1} \equiv p_j \equiv 11 \pmod{30}$ gives $p_{j+1} \equiv 2p_j - 11 \pmod{30}$; the only such number below $2p_j + 16$ and above t_j is at most $2p_j - 11$, so $p_{j+1} \le 2p_j - 11 < r_{i-1}$. By $(*)$ all four members of $\mathcal{Q}_{j+1}$ were subtracted before m_i; in particular two members of $\mathcal{Q}_{j+1}$ were, the first of them at a step before i. Since $j+1 \ge 4$, the same argument applies to $\mathcal{Q}_{j+1}$ and produces two members of $\mathcal{Q}_{j+2}$ subtracted earlier still, and so on without end, which is impossible, the steps being finitely many. Hence (i).

(ii) Let $m_i \in \mathcal{Q}_j$, $j \ge 4$, with $r = r_{i-1}$. If $r \ge p_{j+1}$ then by $(*)$ two members of $\mathcal{Q}_{j+1}$ were subtracted, against (i). So $r < p_{j+1} < 2t_j = 2p_j + 16 \le 2m_i + 16$, hence $m_i > r/2 - 8$ and $r_i = r - m_i < r/2 + 8$.

(iii) Let the members of $\mathcal{Q}_4, \mathcal{Q}_5, \ldots$ be subtracted at the steps $i_1 < \cdots < i_s$, with remainders $\rho_u = r_{i_u - 1}$ before them. By (ii) and the monotonicity of the remainders, $\rho_{u+1} \le r_{i_u} < \rho_u/2 + 8$, that is $\rho_{u+1} + 16 < (\rho_u + 16)/2$, so $\rho_s + 16 < (\rho_1 + 16)/2^{s-1} \le (D + 16)/2^{s-1}$. A member of $\mathcal{Q}_4$ or later was subtracted at step i_s, so $\rho_s \ge 821$ and $837 < (D + 16)/2^{s-1}$, which is the bound on s. The members of $\mathcal{Q}_1, \mathcal{Q}_2, \mathcal{Q}_3$ are twelve, and for $D \ge 821$ one has $(D + 16)/837 \le D/821$, so the total is below $\log_2 D - \log_2 821 + 13 < \log_2 D + 4$. □

The stored equations of the utility follow this process except where a closing fails and a smaller member is tried (Definition 5), and the data agree with the proposition for them as well: below 10^{10} no stored equation uses two members of one quadruplet beyond $\mathcal{Q}_3$, and the only quadruplets used whole are $\mathcal{Q}_0$, $\mathcal{Q}_1$ and $\mathcal{Q}_3$, exactly the two at which the doubling fails together with the royal quad. The proposition does not say that the process ends at 0; what it says is that the doubling, a statement about where the quadruplets lie, bounds the length of the largest-first representation of every integer by $\log_2 D + 4$, while the data put the bound for gaps at eight.

Remark 4. *Coverage is a statement about* all *integers with subtraction allowed and does not imply Conjecture 1, which forbids subtraction, fixes the base m_k and, in its bounded forms, limits the number of terms. It does show that the differences D of the conjecture, which are tiny compared with T_{k-1}, always have* some *derivation of this looser kind.*

The utility prints a derivation of any integer with `-d`, taking the largest quad prime below the remainder first, each prime once, and closing in the vocabulary of the chain equations whenever the choice of primes allows it: royal members added if that suffices, else one product of two members, else two, the primes being re-chosen between those passes. Only when no choice of primes closes with + and × does a minus enter, and only when even that fails a product of three. A range `-d 53 83` prints one line per integer:

```
python3 lc.py -d 16      16 = 13 + 3
python3 lc.py -d 53      53 = 19 + 17 + 7--
python3 lc.py -d 55      55 = 19 + 17 + 11 + 5 + 3
python3 lc.py -d 76      76 = 19 + 17 + (5*7) + 3 + 2
python3 lc.py -d 99      99 = 19 + 13 + (2*5*7) - 3
python3 lc.py -d 101     101 = 19-- + (5*7) + (2*3)
python3 lc.py -d 999986171
                         999986171 = 999900529 + 82729 + 2089 + 821 + 3
```

The greedy choice is not the only derivation; $53 = 19 + 17 + 13 + 7 - 3$ and $55 = 19 + 11 + (7 \times 5)$ are equally valid. Below 300,000 exactly nine integers need a minus at all — 0, 1, 4, 92, 93, 94, 96, 99 and 102, each with only $\mathcal{Q}_1$ or $\mathcal{Q}_1 \cup \{101\}$ to draw on — and one, 99, a product of three royal members.

5 Worked examples

The lines below are copied verbatim from the utility (`python3 lc.py 12 -AME`); `A`, `M`, `M1`, `E1`, `E2`, `E3` label the ways, and a line is omitted when it repeats an earlier line of the same member

(`E2` when it coincides with `E1`; in $\mathcal{Q}_1$, which has no base, `M1` coincides with `M`). Products are written in parentheses and b `^` e means b^e.

$\mathcal{Q}_1 = (11, 13, 17, 19)$ **from the royal quad.** The first quadruplet is written with royal expressions only; the utility also lists alternatives, including ones with subtraction, which the chain equations of Section 3 otherwise never use.

```
11 = (2*3) + 5        also: (2*7) - 3  |  (3*7) - (2*5)  |  7 + 5 + 2 - 3
13 = (2*5) + 3
17 = 7--
19 = (2*7) + 5
```

$\mathcal{Q}_2 = (101, 103, 107, 109)$**: the gap** 82**.** Base $m_2 = 19$, gap $101 - 19 = 82$. The pool is $\mathcal{Q}_1$, whose members other than the base sum to 41, so a royal value is unavoidable; 41 itself is flat, $(5 \times 7) + (2 \times 3)$.

```
A:  101 = 19-- + (5*7) + (2*3)
M:  101 = 19 + (17*3) + (13*2) + 5
M1: 101 = (19*5) + (3*2)
E1: 101 = 19 + (7^2) + (11*3)
E3: 101 = (7^2) + 19 + 17 + 13 + 3
```

$\mathcal{Q}_3 = (191, 193, 197, 199)$**: the same gap with a larger pool.** $191-109 = 82$ is the gap of $\mathcal{Q}_2$ again, but the base is now 109 and 19 is free, so the rule takes it first: $82 = 19+17+13+11+22$ with the flat royal $22 = (3 \times 5) + 7$, one product where $\mathcal{Q}_2$ needed two; no choice of members leaves a plain royal remainder, so pass (ii) is the first to succeed. The utility stores one equation per distinct gap, searched over every member below the gap; a member whose base lies *below* its own gap, as at $\mathcal{Q}_2$ and $\mathcal{Q}_4$, cannot use that base in any of its ways and keeps equations of its own instead. Below 10^9 only two first members do, 101 and 821.

```
A:  191 = 109 + 19-- + (3*5) + 7
M:  191 = 109 + (19*3) + (7*2) + 11
M1: 191 = (19*7) + (17*3) + 5 + 2
E1: 191 = 109 + (7^2) + (11*3)
E3: 191 = (13^2) + (5*3) + 7
```

$\mathcal{Q}_4 = (821, 823, 827, 829)$**: the largest members first.** Gap $821 - 199 = 622 = 197 + 193 + 191 + 19 + 17 + 5$; the three remaining members of $\mathcal{Q}_3$ are taken in turn, then 19 and 17 from $\mathcal{Q}_1$, and the remainder 5, which no unused member fits, is royal. The closing is a plain royal value, so pass (i) succeeds at once and this largest-first form is stored, although a shorter pure sum exists ($197 + 193 + 191 + 17 + 13 + 11$, six members): the rule never looks for the fewest terms, only for a plain closing. The two exponential variants differ here: E1 starts from the largest base 19^2, E2 also, but then prefers $3^5 = 243$ over $5^3 = 125$.

```
A:  821 = 199-- + 19 + 17 + 5
M:  821 = 199 + (197*3) + (13*2) + 5
M1: 821 = (199*3) + (103*2) + 13 + 5
E1: 821 = 199 + (19^2) + (5^3) + 101 + 17 + 11 + 7
E2: 821 = 199 + (19^2) + (3^5) + 11 + 7
E3: 821 = (19^2) + (7^3) + 101 + 11 + 5
```

$\mathcal{Q}_6 = (1871, \ldots)$**: a long gap early in the chain.** Gap $1871 - 1489 = 382$ is one of the four exceptions: no subset of $\mathcal{Q}_1 \cup \mathcal{Q}_2$ sums to 382, so a royal value is needed. The shortest representation with a royal *addition* value has seven terms ($199 + 109 + 19 + 17 + 13 + 11 + 14$, with $14 = 7 + 5 + 2$); with one royal product five terms suffice, $382 = 199 + 109 + 19 + 17 + 38$ with $38 = (5 \times 7) + 3$. The largest-first rule gives exactly this seven-term form, and its closing $14 = 7 + 5 + 2$ is a plain sum, so pass (i) stores it. The multiplicative way is one product.

```
A:  1871 = 1489 + 199 + 109 + 19-- + 7 + 5 + 2
M:  1871 = 1489 + (191*2)
M1: 1871 = (829*2) + (19*7) + (17*3) + 13 + 11 + 5
E1: 1871 = 1489 + (19^2) + (7*3)
E3: 1871 = (11^3) + 199 + 197 + 109 + 17 + 13 + 5
```

$\mathcal{Q}_{12} = (13001, \dots)$**: a two-term gap, and how the rule finds it.** Gap $13001 - 9439 = 3562 = 3461 + 101$, one member of $\mathcal{Q}_9$ and one of $\mathcal{Q}_2$, both first members as the congruence $22 \equiv 11 + 11 \pmod{30}$ requires. About one gap in four has such a two-term representation. The largest-first search takes 3467, the largest member below 3562, and the remainder 95 then needs $19 + 17 + 13 + 11$ and the product (5×7), which pass (i) does not allow; no other choice below 3467 helps, nor below 3463, so the search backs up to 3461, whose remainder 101 is itself a member, and closes at 0.

```
A:  13001 = 9439 + 3461 + 101
M:  13001 = 9439 + (1489*2) + (193*3) + 5
M1: 13001 = (5659*2) + (199*7) + (19*13) + (5*3) + 17 + 11
E1: 13001 = 9439 + (19^2) + (13^3) + (109*7) + 107 + 101 + 17 + 11 + 5
E2: 13001 = 9439 + (13^3) + (19^2) + (109*7) + 107 + 101 + 17 + 11 + 5
E3: 13001 = (109^2) + 829 + 193 + 19-- + (7*5) + 3
```

Two consecutive quadruplets near 3.5×10^7**.** $\mathcal{Q}_{2208}$ and $\mathcal{Q}_{2209}$ are 6,000 apart; the gap of the first, $35545421 - 35524459 = 20962$, is $19429 + 1489 + 19 + 17 + 5 + 3$, closed by the plain royal $8 = 5 + 3$; the gap of the second, $35551421 - 35545429 = 5992$, would close with the product (2×3) after $5659 + 199 + 109 + 19$, so the search backs up from 19 to 17 and closes with $8 = 5 + 3$ instead: $5659 + 199 + 109 + 17 + 5 + 3$.

```
A:  35545421 = 35524459 + 19429 + 1489 + 19 + 17 + 5 + 3
M:  35545421 = 35524459 + (9439*2) + (199*7) + (193*3) + (19*5) + 17
M1: 35545421 = (17771269*2) + (829*3) + (19*13) + (7*5) + 103 + 11
E1: 35545421 = 35524459 + (109^2) + (19^3) + (199*7) + (107*5) + 193 + 101
E2: 35545421 = 35524459 + (7^5) + (13^3) + (19^2) + (107*11) + 199 + 103 + 101 + 17
E3: 35545421 = (5659^2) + (19^5) + (101^3) + 13009 + 1489 + 109 + 103 + 17 + 13

A:  35551421 = 35545429 + 5659 + 199 + 109 + 17 + 5 + 3
M:  35551421 = 35545429 + (2089*2) + (199*7) + (103*3) + (19*5) + 17
M1: 35551421 = (17773369*2) + (1489*3) + (19*11) + 7
E1: 35551421 = 35545429 + (19^2) + (17^3) + (103*5) + (13*7) + 101 + 11
E3: 35551421 = (5659^2) + (19^5) + (101^3) + 19423 + 823 + 193 + 191 + 103 + 7
```

The last quadruplet below 10^9 is $\mathcal{Q}_{28387} = (999986171, \dots)$ with $999986171 = 999900529 + 82729 + 2089 + 821 + 3$, and the largest gap between consecutive quadruplets in that range, 395,520 at $\mathcal{Q}_{19558}$ (gap $D = 395{,}512$), still has a two-term representation, $395512 = 392261 + 3251$; the stored equation, largest first with an added royal 3, is $628641701 = 628246189 + 394819 + 199 + 197 + 193 + 101 + 3$.

6 Computational evidence

The utility. `lc.py` is a single Python 3 file with no dependencies. It finds quadruplets with a segmented sieve (only $p \equiv 11 \pmod{30}$ is tested), then writes every first member's gap in its stored form (Definition 5): largest first, backing up when a choice leads nowhere, the closing tried in the three passes of the derive switch (the minimal lengths reported below come from a separate iterative-deepening search); and stores one set of A/M/E1/E2 equations per distinct gap, plus the base-free M1 and E3 equations of every first member, in a JSON cache that later runs reuse; the other three members of a quadruplet are recorded in their fixed form only. The additive search is capped at 16 terms and 300,000 nodes per call, the M/E searches at 50,000

nodes, the base-free M1 search, whose pool is the whole chain, at 1,000,000 nodes, and the additive remainder of E3 at 300,000 nodes; a fallback path exists for the case that a search gives up, and it was never taken in the data reported here (zero fallback flags). Building the chain to 10^9 took 35 s on a 2019 Intel laptop under Python 3.13.5; the base-free M1 and E3 equations, computed once per first member over the whole chain, account for most of it.

The checker. `check.py` re-reads the JSON and, independently of the search code, verifies every stored equation: the arithmetic, that all members are distinct, that the base is never reused as a term, that every term is smaller than the target, that every royal expression uses each of $2, 3, 5, 7$ at most once and multiplies at most two of them, and that the second, third and fourth member of every quadruplet are stored in their fixed form, and that every stored additive equation is the one Definition 5 prescribes, re-deriving it with a search of its own: the largest-first search with the same three closing passes, the stored equation having to be the result of the first pass that succeeds. On the 10^9 dataset it checked 170,340 equations (6 per first member, the four members of $\mathcal{Q}_1$ included), 3,421 royal expressions, 85,158 fixed forms and 3,411 re-derived additive equations (3,409 closed by 0 or a plain royal sum, 2 needing a product), and found 0 errors.

Digit-sum verification. The independent checker also tests $p \equiv 11 \pmod{30}$, which carries the last digit 1 and $p \equiv 2 \pmod 3$ with it, and $s_{10}(p) \equiv 2 \pmod 3$ for every stored first member. All 28,387 first members below 10^9, from 11 to 999,986,171, satisfy both conditions, with zero violations. Their decimal digit sums range from 2 to 65. The released CSV was checked separately and gives the same count and range. This finite verification illustrates the proposition above; its proof applies beyond the dataset as well.

Minimal lengths. `stats.py` recomputes, for every distinct gap and independently of the preferences of the utility, the minimal length K of a representation under three vocabularies: pool members only (*pure*); pool members plus at most one value of $\mathcal{R}_+$; pool members plus at most one value of $\mathcal{R}$. The minimum is taken when the gap is first met, using only the quadruplets available at that moment. Table 2 gives the result.

Table 2: Minimal length K of a representation of the gap, over the 3,409 distinct first-member gaps met below 10^9.

vocabulary	K=1	2	3	4	5	6	7	8	none	mean
pure members	–	687	–	2608	–	104	–	6	4	3.66
+ one value of $\mathcal{R}_+$	–	688	429	2179	101	5	6	–	1	3.51
+ one value of $\mathcal{R}$	1	705	425	2249	27	2	–	–	0	3.47

The pure column has only even lengths, as it must. The four gaps without a pure representation are those of Conjecture 2; with $\mathcal{R}_+$ only 82 remains; with $\mathcal{R}$ nothing remains and six terms always suffice, which is Conjecture 3. The mean minimal length does not grow along the chain: in eight consecutive blocks of about 3,550 quadruplets the mean pure K of the first member's gap is $3.80, 3.75, 3.73, 3.71, 3.71, 3.67, 3.73, 3.71$ and the share of two-term gaps stays between 22% and 24%.

What the utility actually stores. Table 3 summarises the additive representations chosen by the utility. Because the rule takes the largest member that fits and leaves only what no member covers to the royal quad, the stored line is usually longer than the minimum: 23,817 of the 28,386 first-member equations close with a plain royal sum, 4,539 are pure sums of pool members, and 30, the occurrences of the gap 82, close with a royal product. The longest stored

equation has twelve terms after the base, the mean 5.7. Of the 28,386 gaps, 24,977 repeat a gap seen earlier in the chain; there are only 3,409 distinct gaps, because consecutive quadruplets are often the same distance apart.

Table 3: Additive representations stored by the utility, 10^9 dataset. Length counts the terms after the base, a royal value counting as one.

	length										royal part
scope	2	4	5	6	7	8	9	10	11	12	none / + / ×
first members	1620	6962	6045	3587	5202	2149	2227	486	83	25	4539 / 23817 / 30

Multiplicative and exponential ways. For the 28,386 first-member gaps, the M representations use 90,661 products and 50,439 plain terms, never more than nine terms; 3,005 of the products multiply two royal members. E1 uses 60,176 powers, 49,728 products and 79,246 plain terms (at most 14 terms); E2 uses 63,213 powers, 34,739 products and 78,861 plain terms (at most 12). E1 and E2 coincide for 9,117 of the 28,386 gaps. These representations are released for further study; we make no claim about them beyond their existence within the search caps. The base-free M1 and powers-first E3 representations were likewise found for every first member and for the four members of $\mathcal{Q}_1$ (only 11, 17 and 19, which are written from the royal quad alone, admit no power chain and fall back to the power-first search) and are stored and verified. Of these 28,390 E3 chains, 4,231 have one power, 8,950 two, 11,636 three and 3,570 four; we do not analyse these representations further here.

7 Is this a property of primes, or of density?

A natural objection to Conjecture 2 is that the chain grows dense enough for representability to be forced by counting alone. By the time we reach $\mathcal{Q}_k$ the pool $\mathcal{P}_k$ holds $4(k-1)$ members, so the number of sums of at most eight distinct members is $\binom{4(k-1)}{8}$, vastly exceeding the range of D. On that reading the conjecture would say nothing about primes. This section separates what the objection explains from what it does not.

What is structural

Two features of the data are not evidence about primes, and we do not claim them as such.

The first is the parity of the term counts. Every representation observed uses an *even* number of pool members — the counts are $2, 4, 6, 8$ and never odd — and this is forced. Every member of every $\mathcal{Q}_j$ is an odd prime, so $D = t - m_k$ is a difference of two odd numbers and hence even; a sum of j odd numbers is congruent to j modulo 2; therefore j is even (Lemma 1(v)). No arithmetic beyond parity is involved.

The second is the bound $K = 8$ itself. To test it we ran the same solver on control chains built to have the same spacing as the real one, differing only in that their members are not prime:

- **jitter** — each quadruplet displaced inside its own local gap, preserving order and local density;
- **uniform** — each first member redrawn uniformly in the interval between its neighbours;
- **shift** — each first member displaced by a multiple of 30, preserving the residue class 11 mod 30, and with it the parity argument above, while destroying primality.

Every control keeps the shape $(q, q+2, q+6, q+8)$, the same counting function and the same base rule $m_k = q_{k-1}^{(9)}$, so any difference is attributable to the arithmetic of the members and not to how many there are. The **shift** control attains the same bound of eight terms and the same all-even distribution. The bound is a consequence of the residue class, not of primality.

What is not

chain	worst gap	exceptional gaps	occurrences	largest
real prime quadruplets	8	**4**	98	**382**
control, jitter (seed 1)	12	390	590	1700
control, jitter (seed 2)	11	103	141	1186
control, uniform (seed 1)	9	246	4455	2416
control, uniform (seed 2)	11	65	4221	1328
control, shift	8	26	733	3052

Table 4: All 28,387 quadruplets of each chain below 10^9, 28,386 first-member gaps, under the base rule of Section 2. "Exceptional gaps" counts distinct values of D admitting no purely additive representation; "occurrences" counts the first members that meet one.

The exceptions are a different matter. The real chain has four, the best control has 26 — more than six times as many — and every control admits exceptions several times larger than the real chain's maximum of 382. The contrast survives every control family we tried, including the one that reproduces the bound and the parity exactly.

Density and residue structure together predict that exceptions should be *rare*. They do not predict that there should be only four, that all four should be small, or that the set should close: no new exceptional gap appears after $\mathcal{Q}_{435}$, while 28,000 further quadruplets add none. That is the part of Conjecture 2 we regard as genuinely arithmetic, and the part we would most like to see explained.

8 What the conjecture says about the next quadruplet

It is tempting to read Conjecture 2 as a recipe: take the largest member of the last known quadruplet, add a short sum of earlier members, and the next quadruplet appears. We want to be precise about what is and is not true here.

What it gives. Every quadruplet after the first carries a *certificate* over its predecessors: a short equation whose terms are earlier chain members, with the base term identifying the previous quadruplet. The chain is therefore self-referential in a strong sense: nothing outside $\mathcal{Q}_0, \ldots, \mathcal{Q}_{k-1}$ is needed to write down $\mathcal{Q}_k$ once it is known. The certificate also turns the search for the next quadruplet into a bounded additive search: if the conjecture holds with K terms, then the first member of $\mathcal{Q}_k$ is of the form $m_k + D$ with $D \equiv 22$, 82, 112, 172 or 202 (mod 210), by Lemma 1(iv), and D a sum of at most K distinct pool members, and the candidate can be tested by four primality tests. The utility does exactly this in reverse, producing the certificate after the sieve has found the quadruplet.

What it does not give. The candidates produced by the additive search are far more numerous than quadruplets: near 10^9 the number of two-term sums of pool members below a typical gap is already in the thousands, and almost every admissible D has some short representation. So the conjecture does not point at the next quadruplet, it only says where it will be found once

it exists, and in practice a sieve remains the faster way to find it. Nor does the conjecture imply that a next quadruplet exists; infinitude of prime quadruplets is a separate, open problem.

Why representations are plentiful. The Bateman–Horn heuristic predicts roughly $C_4\, y/(\log y)^4$ quadruplets below y, and so four times as many pool members. The number of j-element subsets of pool members below D grows like

$$\frac{1}{j!}\left(\frac{4C_4\, D}{(\log D)^4}\right)^j,$$

while their sums range over $O(D)$ even numbers, so for $j \geq 2$ the expected number of j-term representations of a given gap grows without bound as D grows, and grows faster the larger j is. This is the reason the mean minimal length stays flat.

Two terms suffice, conditionally. The heuristic can be made precise for two terms. Call an even integer n *admissible* if $n \equiv 22, 82, 112, 172$ or $202 \pmod{210}$, the five classes of Lemma 1(iv); every gap is admissible. Write $r(n)$ for the number of first members $p < n$ such that $n - p$ is also a first member, that is, the number of p for which the eight numbers

$$p,\ p+2,\ p+6,\ p+8, \qquad n-p,\ n-p+2,\ n-p+6,\ n-p+8$$

are all prime. For a fixed n this is a prime-tuple problem for eight linear forms in p, of exactly the kind Hardy and Littlewood considered for Goldbach's problem [1], where the forms are p and $n - p$, and the Bateman–Horn heuristic [2] predicts

$$r(n) \ \sim\ \mathfrak{S}(n)\,\frac{n}{(\log n)^8}, \qquad \mathfrak{S}(n) = \prod_{\ell}\frac{1-\nu_n(\ell)/\ell}{(1-1/\ell)^8},$$

where $\nu_n(\ell)$ is the number of residues $p \bmod \ell$ at which one of the eight forms vanishes. The forms vary with n, so this is a uniform version of the conjecture, as in Goldbach's problem. It is the case of Conjecture 1.2 of Green and Tao [8] for these eight forms, with the Archimedean factor written out as the integral

$$I(n) = \int_{11}^{n-11} \frac{dt}{(\log t)^4\,(\log(n-t))^4}, \qquad r(n) \sim \mathfrak{S}(n)\, I(n),$$

so that $n/(\log n)^8$ above is $I(n)$ to leading order. The system has infinite complexity in their sense, so the conjecture is not a theorem here; it is unproved, and we assume it in the following weak form.

Hypothesis 1. *There are constants $c > 0$ and n_0 such that $r(n) \geq c\,\mathfrak{S}(n)\, n/(\log n)^8$ for every admissible $n \geq n_0$.*

Lemma 3. *$\mathfrak{S}(n) > 0$ if and only if n is admissible, and there is a constant $c_0 > 0$ with $\mathfrak{S}(n) \geq c_0$ for every admissible n.*

Proof. Modulo 2 the four forms $p+\delta$ share the parity of p and the four forms $n-p+\delta$ that of $n-p$, so $\nu_n(2) < 2$ forces n even. Modulo 3 and 5 the first four forms force $p \equiv 2 \pmod 3$ and $p \equiv 1 \pmod 5$, as in the proof of Lemma 1(i), and the last four force the same of $n - p$; hence $n \equiv 22 \pmod{30}$. Modulo 7 the first four force $p \equiv 2, 3, 4$ and the last four $n - p \equiv 2, 3, 4$ (Lemma 1(ii)), hence $n \equiv 4, 5, 6, 0, 1 \pmod 7$, which together with $n \equiv 22 \pmod{30}$ gives exactly the five admissible classes; in the two other classes over $22 \pmod{30}$, $n \equiv 52, 142 \pmod{210}$, some prime $\ell \leq 7$ has $\nu_n(\ell) = \ell$ and $\mathfrak{S}(n) = 0$. For $\ell \geq 11$ at most eight residues are excluded, so $\nu_n(\ell) \leq 8 < \ell$ and the local factor is at least $(1 - 8/\ell)/(1 - 1/\ell)^8 = 1 - 28\,\ell^{-2} + O(\ell^{-3})$, whose product over $\ell \geq 11$ converges to a positive number c_1. On admissible n the factors at $2, 3, 5, 7$ take finitely many values, all positive, with minimum $c_2 > 0$; so $\mathfrak{S}(n) \geq c_1 c_2 =: c_0$. □

Theorem 1. *Assume Hypothesis 1. Then every sufficiently large admissible n is a sum of two distinct first members of prime quadruplets. In particular, every sufficiently large gap $D = p_k - m_k$ is a sum of two first members of earlier quadruplets, and Conjecture 2 holds with $K = 2$ beyond some point.*

Proof. By the hypothesis and Lemma 3, $r(n) \geq c\, c_0\, n/(\log n)^8 \to \infty$, so $r(n) \geq 2$ for all large admissible n. Among the $r(n)$ representations $n = p + (n-p)$ at most one has $p = n - p$, so one has distinct summands. For a gap $D = p_k - m_k$ both summands are below $D < p_k$, hence are first members of quadruplets earlier than $\mathcal{Q}_k$, that is, members of $\mathcal{P}_k$, and the representation is one that Conjecture 2 allows. □

Remark 5. *The theorem is a limiting statement and cannot be made unconditional: it would imply that there are infinitely many prime quadruplets. The two-term share of about 23% reported in Section 6 belongs to the pre-asymptotic range, where the prediction $\mathfrak{S}(n)I(n)$ is of order one; the next paragraph tests that prediction directly.*

A numerical test of the prediction. Hypothesis 1 can be checked where it is of order one. For every admissible $n < 10^7$ we computed $r(n)$ directly from the 898 first members below 10^7, and the prediction $\mathfrak{S}(n)\, I(n)$ as follows (`hltest.py`, two minutes, standard library only). For $\ell \geq 11$ exactly eight residues are excluded unless ℓ divides $n + a + b$ for some offsets $a, b \in \{0, 2, 6, 8\}$, so

$$\mathfrak{S}(n) = L(n)\, G \prod_{\substack{\ell \geq 11 \\ \ell \mid n+a+b}} \frac{\ell - \nu_n(\ell)}{\ell - 8}, \qquad G = \prod_{\ell \geq 11} \frac{1 - 8/\ell}{(1 - 1/\ell)^8} = 0.23242,$$

where $L(n)$ is the exact factor at $2, 3, 5, 7$, equal to 639.15, 1278.31, 1278.31, 639.15 and 1917.46 on the classes $22, 82, 112, 172, 202 \pmod{210}$, and the finite product runs over the prime divisors of the nine numbers $n, n+2, \ldots, n+16$. The integral $I(n)$ was evaluated by Simpson's rule in $\log t$. Table 5 sums both sides over the admissible n of each decade and also compares the share of n with $r(n) = 0$ with the Poisson value $\exp(-\mathfrak{S}(n)I(n)/2)$, the factor $\frac{1}{2}$ because r counts each unordered pair twice.

The raw ratio runs from 0.79 to 1.12. This is the slow convergence of the quadruplet count itself: the same constants give 4.1512 for C_4, and the number of first members below x stands at 0.863, 0.893, 0.958 and 1.055 of $C_4 \int_{11}^{x} dt/(\log t)^4$ at $x = 10^4, \ldots, 10^7$. A two-term count involves two first members, so the prediction inherits roughly the square of that deviation; dividing the ratio of each decade by the square of the single-quadruplet ratio at its upper end gives 1.11, 0.99, 0.96 and 1.01. The binary correlation that Conjecture 1.2 adds to the single-quadruplet count, the factor $\mathfrak{S}(n)/C_4^2$ with its dependence on the prime divisors of $n + a + b$, is therefore confirmed to within a few percent at 10^4 to 10^7, for a system of infinite complexity. The Poisson shares agree in the same way.

For the gaps of the chain the same prediction gives the expected number of gaps with a two-term pure form, $\sum_D (1 - e^{-\mathfrak{S}(D)I(D)/2})$: 8,084 against 6,475 observed below 10^9 and 52,660 against 41,684 below 10^{10}, ratios 0.80 and 0.79, as expected for gaps that mostly lie between 10^4 and 10^5. The four royal gaps are not singled out by this prediction: at 82, 172 and 382 it gives 1.48, 0.65 and 0.58, in line with the other twenty gap values below 1000, of which 18 have no two-term form either. What distinguishes them is that at these sizes the four-, six- and eight-term forms are missing too, because the pool is $\mathcal{Q}_1 \cup \mathcal{Q}_2$ and nothing more; from $\mathcal{Q}_3$ on the longer forms take over, and the two-term forms that the theorem speaks of become the rule only far beyond the range of this note.

Question 1. *Is every sufficiently large gap a sum of* two *distinct earlier members unconditionally, that is, do the lengths $4, 6, 8$ in Table 2 eventually disappear? Theorem 1 says yes under*

Table 5: Two-term representations by first members against the prediction of Conjecture 1.2 of [8] (Hypothesis 1 with the asymptotic $\mathfrak{S}(n)I(n)$), summed over the admissible n of each decade below 10^7. The column δ is the ratio of the number of first members below the upper end of the decade to $C_4 \int_{11}^{x} dt/(\log t)^4$; the last two columns give the share of admissible n with no two-term representation and its Poisson prediction.

n	admissible n	$\sum r(n)$	$\sum \mathfrak{S}(n)I(n)$	ratio	δ	ratio/δ^2	$r = 0$	Poisson
$[10^3, 10^4)$	215	92	112	0.82	0.863	1.11	0.786	0.779
$[10^4, 10^5)$	2,143	925	1,169	0.79	0.893	0.99	0.803	0.770
$[10^5, 10^6)$	21,428	17,198	19,447	0.88	0.958	0.96	0.681	0.656
$[10^6, 10^7]$	214,286	512,765	457,133	1.12	1.055	1.01	0.363	0.406

Hypothesis 1, and Lemma 3 shows that there is no local obstruction: of the seven classes modulo 210 *that lie over* 22 (mod 30)*, the two that contain no sum of two first members,* 52 *and* 142*, are exactly the two that Lemma 1(iv) excludes for gaps.*

Question 2. *Is a royal value ever needed again beyond the four exceptional gaps, i.e. is Conjecture 2 true with exactly these exceptions and $K = 8$, or even $K = 6$ beyond some point?*

Question 3. *The same construction applies to any prime constellation, for instance twin primes or sexy prime triplets, with the corresponding admissible residues. Which constellations admit a bounded chain, and with what K?*

Question 4. *The exceptional gaps are exactly the coverage holes of $\mathcal{Q}_1 \cup \mathcal{Q}_2$. Is there a simple description of the set of gaps that are not sums of pool members at the moment they are first met, for the general constellation?*

9 Availability and reproducibility

The utility, the checker and the statistics script are released in the folder `LC` of the repository

https://github.com/SPAlgorithm/LE

The utility, the dataset, the Zenodo record and the companion app refer to the conjecture as *Ladhe's Quad Conjecture*; this note uses the neutral name. The commands below rebuild everything reported here from scratch.

```
python3 lc.py --upto 999999999 --cache paper/qarray_1e9.json   # 35 s
python3 paper/check.py paper/qarray_1e9.json                   # 0 bad
python3 paper/stats.py paper/qarray_1e9.json                   # all figures
python3 lc.py 25 -AME --cache paper/qarray_1e9.json            # the examples
python3 paper/residues.py paper/quads_1e9.csv      # Lemma 1, 4 s
python3 paper/density.py 28387                     # Section 7, ~5 min
python3 paper/hltest.py --upto 10000000            # Section 8, Table 5, ~2 min
```

The dataset `qarray_1e9.json` (23,147,955 bytes) has SHA-256

`af4306fc1dfb616592366fc04b064570caa65c31444faa4fbe82406b49b36a89`

and `quads_1e9.csv` has one row per member (113,548 rows, $\mathcal{Q}_1$ included): for a first member the base, the gap, the four A/M/E1/E2 equations (the base-free M1/E3 equations are in the JSON only) and the three minimal lengths; for the other three members of a quadruplet only their fixed form. The control chains of Section 7 are generated inside `density.py` from a fixed seed, so its table is reproducible without any further data. Interactive use: `python3 lc.py` asks for a number of quadruplets; `-M`, `-E`, `-AME` select the ways; `--all` lists all four members in each heading; `--full` spells out the members behind a `19--`; `-v` shows which quadruplet each term comes from; `-d N` prints a derivation of any integer N as in Section 4, extending the chain first if N lies beyond the cache.

Videos. Two narrated explanations accompany this note, under the name the utility and the app use: *Ladhe's Quad Conjecture: A New Pattern in Prime Quadruplets* (28 min, `https://youtu.be/_XSOJ0Yj77Q`), which covers Sections 3–8, and *Build Any Number From Prime Quadruplets* (11 min, `https://youtu.be/JNWvt3hS540`), which walks through the coverage proposition of Section 4 by hand.

App. The authors' free iOS app *LE-Games* contains an interactive companion to this note: the chain itself with every way of writing each first member, a matching game (*QuadQuest*) whose every board is an equation of the chain, and a printable practice sheet: `https://apps.apple.com/app/id6767880385`.

Use of AI tools. The authors prepared this paper with substantial assistance from Claude (Anthropic), a large language model, and state its role here in full. The observation, the choice of rules, the conjecture and the decision of what to compute and publish are the authors'. The utility `lc.py`, the checker `check.py` and the statistics scripts were written with the model's assistance; the dataset was produced by the utility and re-verified by the independent checker. The prose of every section was drafted and edited with the model, and the proofs of Lemma 1, Proposition 1, Proposition 2 and Theorem 1 were drafted with its help and then checked line by line by the authors, who take full responsibility for every statement, equation and number in the paper. No AI tool is an author.

Conflict of interest. The authors declare no competing interests related to this note.

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A The first 25 quadruplets

For the first member t of each quadruplet: the quad index k, the base m_k, the gap D, the additive equation stored by the utility (base first; a fully used quadruplet folded to `19--` as in Section 2) with, under it, the shortest *no-royals* form of the same member (base plus pool members only, the larger members first among forms of the same length; `none` for the four exceptional gaps), the length of the stored equation including the base, the minimal pure length K of the gap ("–" for the exceptional gaps), and, when the gap was met before, the quad where its equation was first found. $\mathcal{Q}_1$ is written from royal members only. The second, third and fourth member of every quadruplet are $p+2$, $p+(2\times 3)$ and $(p+6)+2$ and are not listed.

k	t	m_k	D	additive equation	len	K	from
1	11			`(2*3) + 5`	2		
2	101	19	82	`19-- + (5*7) + (2*3)`	6	–	
				no royals: `none`			
3	191	109	82	`109 + 19-- + (3*5) + 7`	7	–	2
				no royals: `none`			
4	821	199	622	`199-- + 19 + 17 + 5`	7	6	
				no royals: `199-- + 17 + 13 + 11`			
5	1481	829	652	`829 + 199 + 197 + 193 + 19-- + 3`	9	8	
				no royals: `829 + 197 + 191 + 103 + 101 + 19--`			
6	1871	1489	382	`1489 + 199 + 109 + 19-- + 7 + 5 + 2`	10	–	
				no royals: `none`			
7	2081	1879	202	`1879 + 199 + 3`	3	2	
				no royals: `1879 + 191 + 11`			
8	3251	2089	1162	`2089 + 829 + 199 + 109 + 17 + 5 + 3`	7	6	
				no royals: `2089 + 829 + 191 + 101 + 17 + 13 + 11`			
9	3461	3259	202	`3259 + 199 + 3`	3	2	7
				no royals: `3259 + 191 + 11`			
10	5651	3469	2182	`3469 + 2081 + 101`	3	2	
				no royals: `3469 + 2081 + 101`			
11	9431	5659	3772	`5659 + 3469 + 199 + 101 + 3`	5	4	
				no royals: `5659 + 3469 + 191 + 101 + 11`			
12	13001	9439	3562	`9439 + 3461 + 101`	3	2	
				no royals: `9439 + 3461 + 101`			

k	t	m_k	D	additive equation	len	K	from
13	15641	13009	2632	`13009 + 2089 + 199 + 197 + 109 + 19 + 17 + 2` *no royals:* `13009 + 1489 + 827 + 191 + 101 + 13 + 11`	8	6	
14	15731	15649	82	`15649 + 19-- + (3*5) + 7` *no royals:* `none`	7	–	2
15	16061	15739	322	`15739 + 199 + 109 + 11 + 3` *no royals:* `15739 + 197 + 101 + 13 + 11`	5	4	
16	18041	16069	1972	`16069 + 1871 + 101` *no royals:* `16069 + 1871 + 101`	3	2	
17	18911	18049	862	`18049 + 829 + 19 + 11 + 3` *no royals:* `18049 + 821 + 17 + 13 + 11`	5	4	
18	19421	18919	502	`18919 + 199 + 197 + 103 + 3` *no royals:* `18919 + 199 + 191 + 101 + 11`	5	4	
19	21011	19429	1582	`19429 + 1481 + 101` *no royals:* `19429 + 1481 + 101`	3	2	
20	22271	21019	1252	`21019 + 829 + 199 + 197 + 19 + 5 + 3` *no royals:* `21019 + 829 + 191 + 107 + 101 + 13 + 11`	7	6	
21	25301	22279	3022	`22279 + 2089 + 829 + 101 + 3` *no royals:* `22279 + 2089 + 821 + 101 + 11`	5	4	
22	31721	25309	6412	`25309 + 5659 + 199 + 197 + 193 + 109 + 19 + 17 + 11 + 5 + 3` *no royals:* `25309 + 3257 + 1483 + 1481 + 191`	11	4	
23	34841	31729	3112	`31729 + 2089 + 829 + 191 + 3` *no royals:* `31729 + 2089 + 821 + 191 + 11`	5	4	
24	43781	34849	8932	`34849 + 5659 + 3259 + 11 + 3` *no royals:* `34849 + 5657 + 3251 + 13 + 11`	5	4	
25	51341	43789	7552	`43789 + 5659 + 1879 + 11 + 3` *no royals:* `43789 + 5657 + 1871 + 13 + 11`	5	4	

B The four royal gaps

Gaps that are not a sum of distinct pool members at the moment they are first met, with the additive (A), multiplicative (M) and exponential (E1) representations of the gap stored by the utility.

D	k	t	A	M	E1
22	169	1006331	`19 + 3`	`(11*2)`	`(3^2) + 13`
82	2	101	`17 + 13 + 11 + (5*7) + (2*3)`	`(17*3) + (13*2) + 5`	`(7^2) + (11*3)`
172	435	3512231	`109 + 19-- + 3`	`(19*7) + (17*2) + 5`	`(13^2) + 3`
382	6	1871	`199 + 109 + 19-- + 7 + 5 + 2`	`(191*2)`	`(19^2) + (7*3)`

C Notation

$\mathcal{Q}_0$	the royal quad $(2, 3, 5, 7)$
$\mathcal{Q}_k$, $k \geq 1$	the k-th prime quadruplet $(p, p+2, p+6, p+8)$, $p \geq 11$
$q_k^{(d)}$	the member of $\mathcal{Q}_k$ ending in digit $d \in \{1, 3, 7, 9\}$
$m_k = q_{k-1}^{(9)}$	the base of $\mathcal{Q}_k$
$D = p_k - m_k$	the gap of the first member of $\mathcal{Q}_k$
$\mathcal{P}_k = \mathcal{Q}_1 \cup \cdots \cup \mathcal{Q}_{k-1}$	the pool available to $\mathcal{Q}_k$
$\mathcal{R}$, $\mathcal{R}_+$	royal values: all (30, every one flat), additive (11)
$\mathcal{R}^{\pm}$	non-negative royal values with $+, -, \times$ (83)
`19--`, `7--`	one whole quadruplet, $19 + 17 + 13 + 11$; the royal quad, $7 + 5 + 3 + 2$
K	minimal pure length of a gap
C_k, T_k	integers with a derivation over $\mathcal{Q}_0..\mathcal{Q}_k$; largest T with $[0, T] \subseteq C_k$
S_k	$S_3 = T_3$, $S_k = S_{k-1} + 4p_k + 16$: the proven lower bound for T_k